

# **Financial Investment Analysis Report**

Done By

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# Comparative Evaluation: Apple (AAPL), Microsoft (MSFT), and S&P 500 Index

## 1. Introduction

The financial markets are dynamic and influenced by various global, economic, and sector-specific factors. To make informed investment decisions, it's critical to analyze stocks from both a return and risk perspective. This project presents a **comprehensive financial analysis** of two leading technology stocks—**Apple Inc. (AAPL)** and **Microsoft Corporation (MSFT)** in comparison to the **S&P 500 Index**, which represents the broader U.S. market.

As a business user, our goal is to **visualize trends**, evaluate **risk-adjusted returns**, and **interpret stock behavior** using historical performance data. The insights generated can guide strategic decisions, portfolio adjustments, or financial reporting.

## 2. Objective

The objective of this report is to conduct a comparative financial investment analysis of Apple Inc. (AAPL) and Microsoft Corporation (MSFT) relative to the S&P 500 index. The analysis aims to evaluate the performance, risk, and return of these two technology stocks over a specified period using key financial metrics such as daily and annual returns, volatility, beta, and  $R^2$ . By leveraging Python and Excel-based data analysis tools, this report provides data-driven insights to support informed investment decision-making and portfolio evaluation strategies.

## 3. Data Source and Structure

- **Data Source:** Yahoo Finance (Pulled using Python's yFinance API)
- **Period:** January 1, 2023 – April 2025 (Approx.)
- **Companies Analyzed:**
  - **AAPL** – Apple Inc.
  - **MSFT** – Microsoft Corporation

- **^GSPC** – S&P 500 Index
- **Tools Used:**
  - Python (for initial cleaning and analysis)
  - Microsoft Excel (for visualization and extended calculations)

#### **Dataset Overview:**

- The data consists of **1699 daily trading records**.
- Each record includes:  
Date, Open, High, Low, Close, Adj Close, Volume

A combined dataset with aligned dates across all tickers ensures comparability in return and risk metrics.

## **4. Data Cleaning and Preparation**

To ensure analytical accuracy and support meaningful insights, several systematic data cleaning and preparation steps were applied to the stock data of **Apple (AAPL)**, **Microsoft (MSFT)**, and the **S&P 500 Index (^GSPC)**. These steps are outlined

### **a. Data Loading**

- Imported an Excel workbook containing separate sheets for each ticker: AAPL, MSFT, and ^GSPC.
- Loaded each sheet into a pandas Data Frame and set the index as the date.

```
import pandas as pd

# Load Excel file
file_path = "Full_Stock_Data_AAPL_MSFT_SP500.xlsx"
xls = pd.read_excel(file_path, sheet_name=None, index_col=0)
```

### **b. Missing Value Removal**

- All rows containing missing values (e.g., from non-trading days or incomplete records) were removed to prevent any distortions in the analysis.

```
# Loop through each sheet/ticker
for ticker, df in xls.items():
    df = df.dropna() # Remove rows with any missing data
    df = df.astype(float) # Convert all columns to float (if not already)
    df['Ticker'] = ticker # Add a column for ticker name
    cleaned_data[ticker] = df
```

- This ensures that all remaining data points are complete and valid across all relevant metrics (Open, High, Low, Close, Adj Close, Volume).

### c. Data Type Consistency

- All numeric columns were explicitly converted to float data types to standardize the format and avoid errors during statistical or visual analysis.

### d. Ticker Tagging

- A new column, '**Ticker**', was added to each dataset to retain source identity after merging datasets.
- This tagging allows a clear differentiation between records from AAPL, MSFT, and ^GSPC.

### e. Dataset Consolidation

- All individual stock datasets were concatenated into a unified **master dataset** for easier cross-ticker comparison and collective analysis.
- The combined dataset includes all cleaned records from the individual tickers.

```

# Optionally combine all into a single sheet
combined_df = pd.concat(cleaned_data.values())

# Save cleaned data to a new Excel file
with pd.ExcelWriter("Cleaned_Stock_Data_AAPL_MSFT_SP500.xlsx") as writer:
    for ticker, df in cleaned_data.items():
        df.to_excel(writer, sheet_name=ticker)
    # Optional: Add combined sheet
    combined_df.to_excel(writer, sheet_name="Combined")

print("Data cleaned and saved to 'Cleaned_Stock_Data_AAPL_MSFT_SP500.xlsx'.")

```

## f. Output Generation

- Cleaned datasets were saved into a new Excel file:  
"Cleaned\_Stock\_Data\_AAPL\_MSFT\_SP500.xlsx".
- This file contains:
  - Separate sheets for each ticker.
  - A consolidated "**Combined**" sheet for integrated analysis.

## g. Summary of Final Dataset Structure

After cleaning, the resulting dataset contains no missing values across any of the following columns:

- Open, High, Low, Close, Adj Close, Volume, Ticker

```
df.head()
```

|            | Open        | High        | Low         | Close       | Adj Close   | Volume       | Ticker |
|------------|-------------|-------------|-------------|-------------|-------------|--------------|--------|
| Date       |             |             |             |             |             |              |        |
| 2023-01-03 | 3853.290039 | 3878.459961 | 3794.330078 | 3824.139893 | 3824.139893 | 3.959140e+09 | ^GSPC  |
| 2023-01-04 | 3840.360107 | 3873.159912 | 3815.770020 | 3852.969971 | 3852.969971 | 4.414080e+09 | ^GSPC  |
| 2023-01-05 | 3839.739990 | 3839.739990 | 3802.419922 | 3808.100098 | 3808.100098 | 3.893450e+09 | ^GSPC  |
| 2023-01-06 | 3823.370117 | 3906.189941 | 3809.560059 | 3895.080078 | 3895.080078 | 3.923560e+09 | ^GSPC  |
| 2023-01-09 | 3910.820068 | 3950.570068 | 3890.419922 | 3892.090088 | 3892.090088 | 4.311770e+09 | ^GSPC  |

This structured, clean dataset now serves as a robust foundation for subsequent **data visualization**, **time series analysis**, and **statistical modeling**.

- Identified **outlier days** in trading volume using the **Interquartile Range (IQR)** method.
- Combined individual sheets (AAPL, MSFT, ^GSPC) into a single master dataset.
- Calculated **daily returns** using Adjusted Close prices.

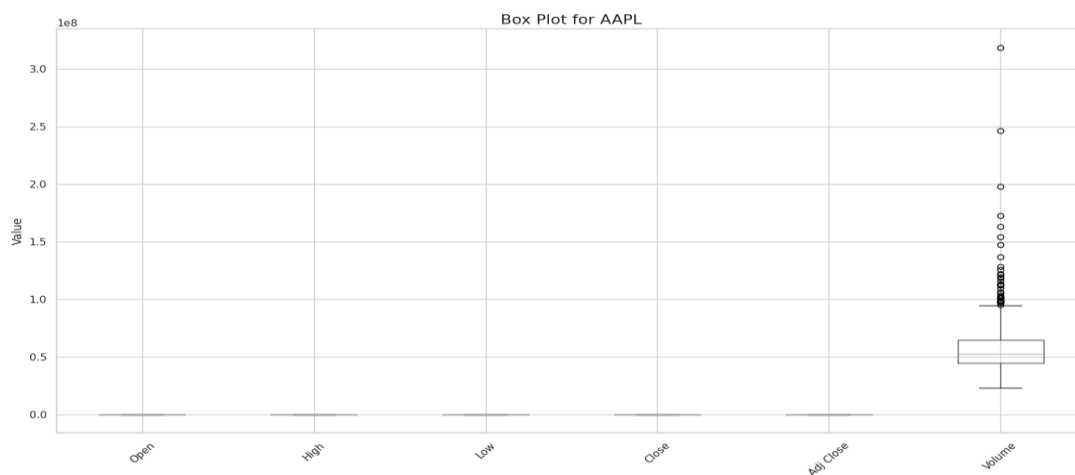
These preparation steps helped build a reliable foundation for further visual and statistical analysis.

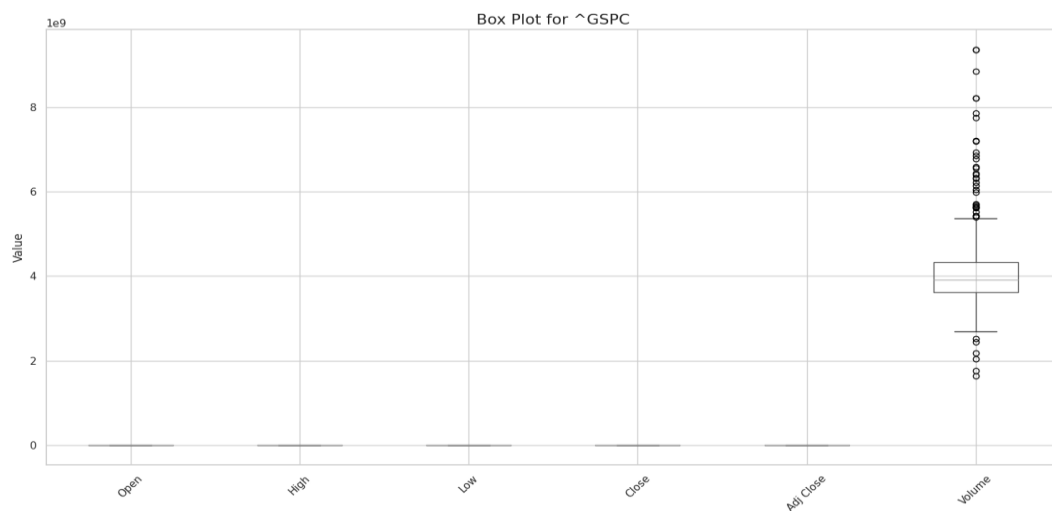
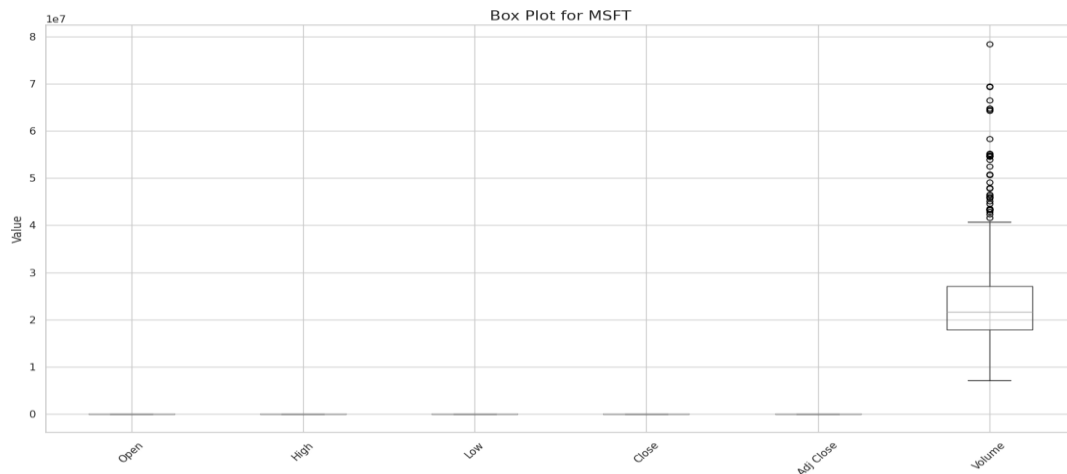
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## 5. Volume Outlier

Identified outlier days in trading volume using the Interquartile Range (IQR) method

Boxplots were created for each stock to visualize spread and outliers in trading volume. These plots clearly show the extreme values that deviate from typical market behavior, indicating institutional reactions or speculative trading days.





### AAPL (Apple Inc.)

- The **Volume** shows a high number of outliers on the upper end.
- This indicates **frequent unusually high trading days**, likely due to earnings releases, product launches (e.g., iPhones), or macroeconomic events.
- The spread between the upper whisker and outliers is wide, suggesting **significant spikes** in trading volume on select days.
- This also reflects **high investor activity or market reaction** to Apple-specific news.

### MSFT (Microsoft Corp.)

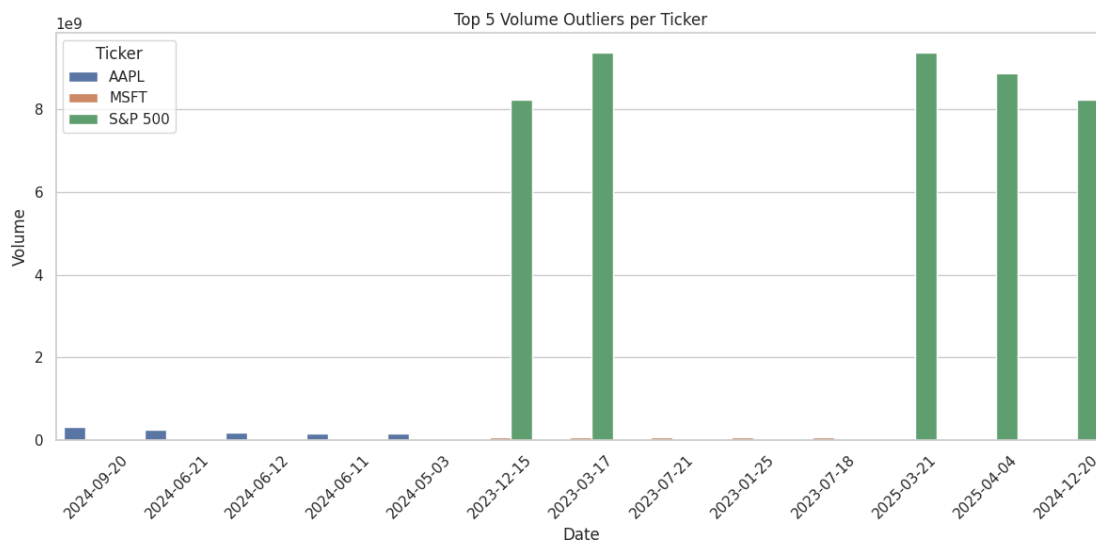
- MSFT's Volume also shows **upper outliers**, though fewer and less extreme compared to AAPL.
- This implies **less volatility in trading activity**, with fewer exceptionally high-volume days.
- Still, the presence of outliers suggests there were **notable events or reactions** affecting the stock intermittently.

### S&P 500 Index (^GSPC)

- Despite being an index, the Volume shows **a large number of significant outliers**.
- This is expected due to **market-wide reactions**—like Fed announcements, geopolitical events, or major economic reports.

Outliers are more extreme and frequent than in MSFT and even AAPL, emphasizing **broader market volatility**.

### Top 5 Outlier Trading Volume Days:



| Ticker  | Volume        | Date       |
|---------|---------------|------------|
| AAPL    | 318,679,900   | 2024-09-20 |
|         | 246,421,400   | 2024-06-21 |
|         | 198,134,300   | 2024-06-12 |
|         | 172,373,300   | 2024-06-11 |
|         | 163,224,100   | 2024-05-03 |
| MSFT    | 78,478,200    | 2023-12-15 |
|         | 69,527,400    | 2023-03-17 |
|         | 69,368,900    | 2023-07-21 |
|         | 66,526,600    | 2023-01-25 |
|         | 64,872,700    | 2023-07-18 |
| S&P 500 | 9,367,460,000 | 2025-03-21 |
|         | 9,354,280,000 | 2023-03-17 |
|         | 8,853,500,000 | 2025-04-04 |
|         | 8,223,220,000 | 2024-12-20 |
|         | 8,218,980,000 | 2023-12-15 |

- **AAPL:** Volume spikes align with major events such as earnings reports, product launches (iPhone, Vision Pro), or quarterly disclosures. June 21, 2024, marks a **triple witching day**, contributing to the surge.

- **MSFT**: High volume dates often relate to **AI partnerships**, cloud segment results, or earnings releases. The December 2023 spike could align with **OpenAI developments** or Microsoft Cloud announcements.
- **S&P 500**: Outliers align with **macroeconomic events** (e.g., Fed announcements, inflation data, geopolitical changes) or **options expirations**, driving large-scale institutional activity.

## 6. Key Financial Calculations Performed

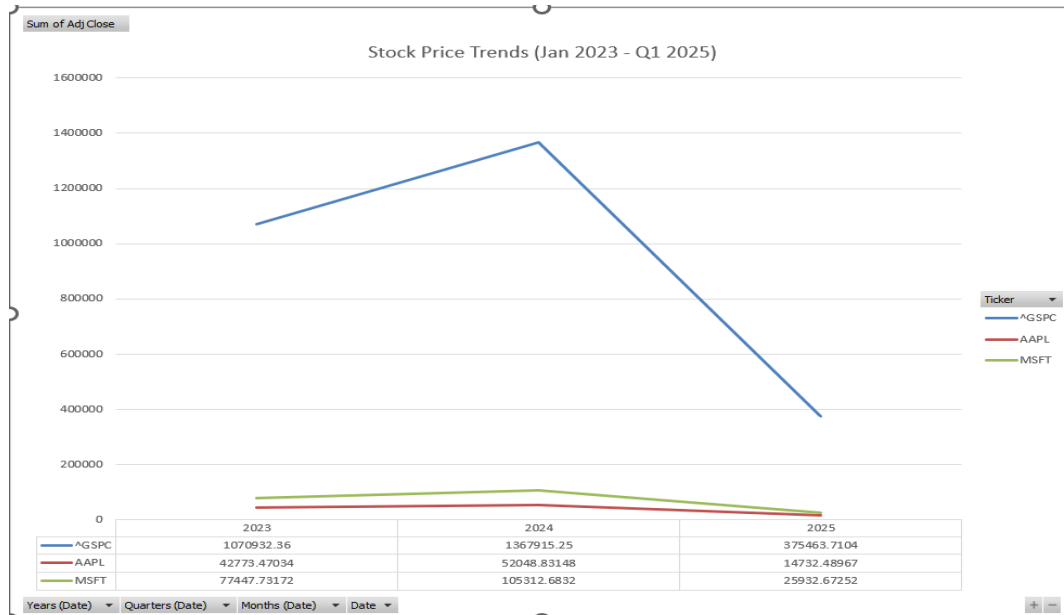
We computed several investment metrics to assess stock performance:

| Metric                | Formula Used                                             |
|-----------------------|----------------------------------------------------------|
| Daily Returns         | $= (\text{Today} - \text{Yesterday}) / \text{Yesterday}$ |
| Annual Return (%)     | $= \text{AVERAGE}(\text{Daily Return}) * 252$            |
| Annual Volatility (%) | $= \text{STDEV}(\text{Daily Return}) * \text{SQRT}(252)$ |
| Beta (vs S&P 500)     | Regression of Stock vs Market returns                    |
| R <sup>2</sup> Value  | Coefficient of determination from regression             |

All formulas were implemented in **Excel**, allowing for transparency and dynamic recalculations based on updated data.

## 7. Visual Explorations and Insights

### A. Price Trend Lines (AAPL, MSFT, S&P 500)



A line chart comparing the Adjusted Close prices over time shows the following

#### Insights from the Stock Price Trend Chart (2023 – Q1 2025)

##### 1. General Growth Trends (2023–2024)

- All three tickers — **AAPL, MSFT, and S&P 500** — showed **consistent growth in adjusted closing prices** from 2023 into 2024.
- This suggests **positive investor sentiment**, possibly due to favorable macroeconomic conditions, strong earnings, or tech-driven optimism.

##### 2. Dip in Q1 2025

- In early 2025, all three lines **decline sharply**, most notably:
  - **S&P 500** experiences a **significant drop**, possibly due to **broad market volatility or correction**.

- **AAPL** and **MSFT** also decline, but with **less intensity**, indicating tech sector may have shown **relative resilience**.

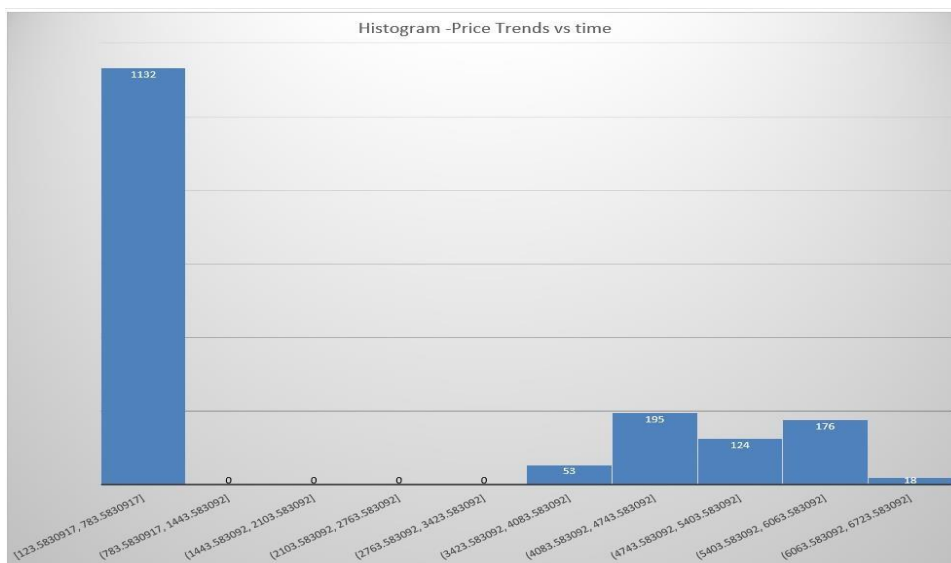
### 3. MSFT vs AAPL Performance

- **MSFT** consistently maintains a **higher adjusted price** than AAPL across all periods.
- This could imply **better growth performance**, stronger fundamentals, or higher investor confidence.

### 4. S&P 500 Volume Overshadowing Individual Stocks

- The chart shows **S&P 500 values in the million range**, dwarfing individual stocks — expected since the index aggregates 500 companies.
  - The S&P 500 served as a benchmark with more **stabilized trends**, reacting to broader market movements but with less volatility.
  - Interpretation should focus more on **trajectory** than raw numbers.
- 

## B. Histogram – Price Trends vs Time



The data suggests that the price was most frequently in the lower range (123.54 to 783.58). There were no occurrences of prices in the mid-ranges. Later time intervals saw an increase in

the frequency of higher price points, with the range of 5403.58 to 6063.58 being the second most frequent. The highest price range (6063.58 to 6723.58) was rarely observed.

Here's what we can learn from this visual:

- **Dominant Price Range:** The first bar is significantly taller than all others, reaching a frequency of 1132. This indicates that the price range spanning approximately 123.54 to 783.58 (based on the x-axis labels) occurred far more frequently than any other price range during the observed period.
- **Infrequent Mid-Range Prices:** The bars representing the price ranges of approximately 783.56 to 1443.58, 1443.58 to 2303.58, and 2303.58 to 2763.58 all have a frequency of 0. This suggests that prices within these ranges were not observed at all during the time period captured by this histogram.
- **Increasing Frequency in Later Time Intervals:** Moving towards the right side of the histogram, we see a non-zero frequency starting with the price range of approximately 3423.58 to 4083.58 (frequency of 53). The frequencies then generally increase, with the next two bars (approximately 4083.58 to 4743.58 and 4743.58 to 5403.58) showing frequencies of 155 and 124 respectively.
- **High Frequency in the Second to Last Range:** The price range of approximately 5403.58 to 6063.58 shows a relatively high frequency of 176, the second highest in the entire distribution.
- **Low Frequency in the Final Range:** The last bar, representing the price range of approximately 6063.58 to 6723.58, has a very low frequency of 18.

C. Annual Return and Volatility (Clustered Bar Chart)

| Summary Table |               |                   |
|---------------|---------------|-------------------|
| Column1       | Annual Return | Annual Volatility |
| AAPL          | -22.97%       | 70.77%            |
| MSFT          | -22.91%       | 70.62%            |
| S&P500        | -30.91%       | 68.20%            |



Annual Return vs Risk (Volatility) – Comparative Analysis of AAPL, MSFT & S&P500 (2023–Q1 2025)"

- a. Performance Overview:
- All three tickers experienced negative annual returns over the analyzed period, suggesting a broader market downturn or correction phase.
  - The **S&P 500** (market benchmark) showed the **worst annual return of -30.91%**, highlighting the overall bearish sentiment in the market.
- b. Company vs Market Performance:
- Both **AAPL (-22.97%)** and **MSFT (-22.91%)** outperformed the **S&P 500**, indicating stronger resilience in individual company stocks compared to the general market.
  - This suggests that while tech stocks dipped, they **sustained less damage** than the broader index — a positive signal for long-term investors.
- c. Risk Analysis (Volatility):
- Surprisingly, **Apple (70.77%)** and **Microsoft (70.62%)** exhibited **higher annualized volatility** than the S&P 500 (68.20%), signaling greater daily price fluctuations.

- Higher volatility often means higher risk, but when paired with better relative returns than the index, it could also suggest **more active investor interest** or **speculative trading behavior**.

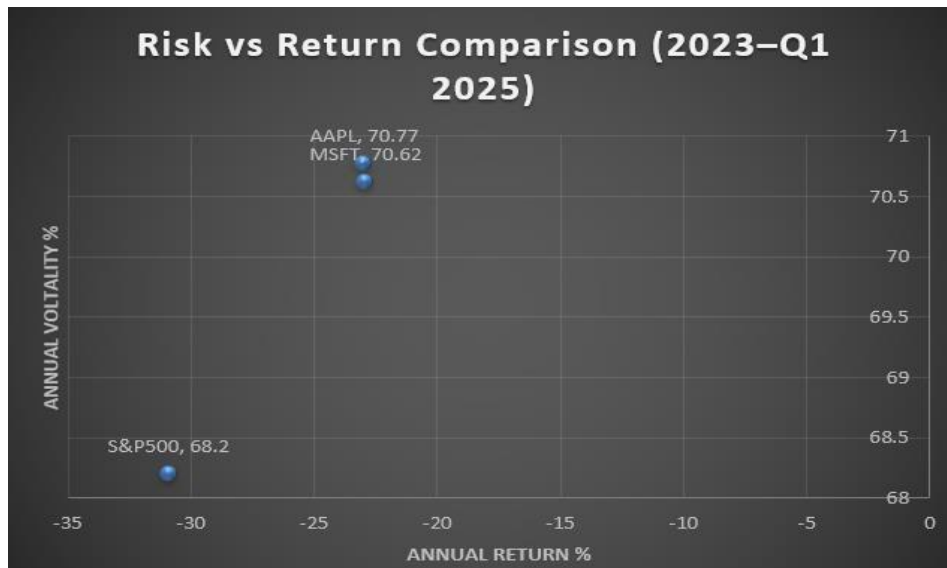
*d. Interpretation of Risk vs Return:*

- The return-risk balance tells us:
  - **AAPL and MSFT carry higher risk**, but also better relative performance than the market.
  - **S&P500 offers lower risk** but performed worse in this period, suggesting a **poor risk-reward trade-off** during this timeframe.

The above chart illustrates the comparative annual return and volatility across Apple (AAPL), Microsoft (MSFT), and the S&P 500 index for the period between January 2023 and Q1 2025. While all three investments yielded negative returns, AAPL and MSFT demonstrated comparatively better performance than the market average, suggesting stronger investor confidence and corporate resilience. Despite showing higher volatility than the S&P 500, these tech giants offered a relatively better risk-adjusted return. This implies that during volatile market conditions, quality tech stocks may serve as a defensive play within an investment portfolio.

#### D. Risk vs Return (Scatter Plot)

| Ticker | Annual Return (%) | Annual Volatility (%) |
|--------|-------------------|-----------------------|
| AAPL   | -22.97            | 70.77                 |
| MSFT   | -22.91            | 70.62                 |
| S&P500 | -30.91            | 68.2                  |



The **Risk vs Return scatter plot** offers a critical view of how each investment, **AAPL (Apple)**, **MSFT (Microsoft)**, and the **S&P 500 index** performed in terms of **reward relative to the level of risk**.

1. **AAPL and MSFT appear close together** on the plot, reflecting very similar risk-return profiles:
  - **Annual Return:** AAPL (-22.97%) vs MSFT (-22.91%)
  - **Annual Volatility:** AAPL (70.77%) vs MSFT (70.62%)

This tight clustering suggests that both stocks **reacted similarly to market conditions**, likely due to their shared sector (technology) and large-cap profile. Despite the negative returns, their **performance per unit of risk** was relatively better than the broader market.

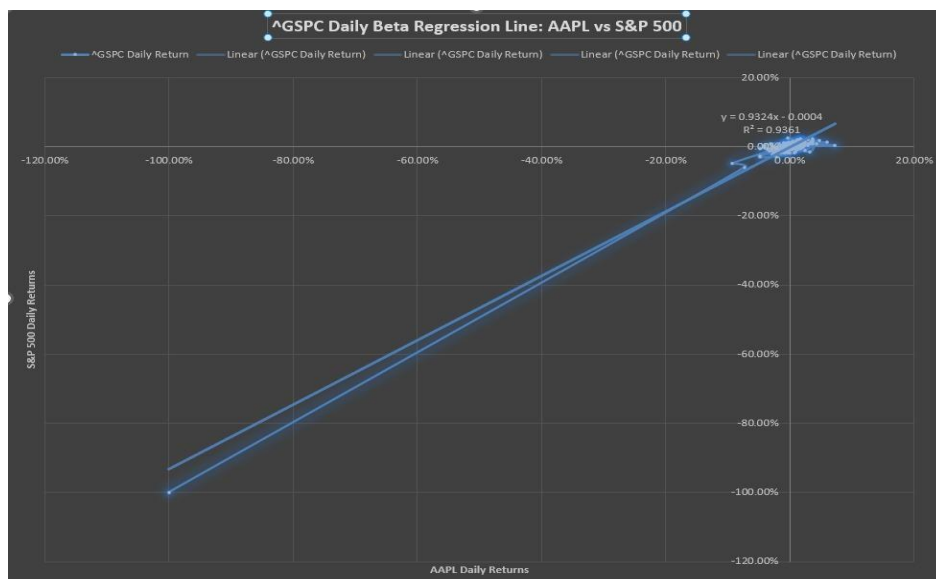
2. **S&P 500 is positioned lower on the return axis** while maintaining slightly lower volatility (68.20%).

- This indicates that the **index experienced the sharpest loss (-30.91%)**, yet it was **not significantly less volatile** than the individual stocks.
- This underperformance relative to its risk indicates a **less efficient risk-return trade-off** during the observed period.

3. The visual clustering of all three in the **high-risk, negative-return quadrant** suggests that:

- The market overall went through a **challenging phase** (possibly influenced by macroeconomic factors like interest rate hikes, inflation concerns, or global instability).
- Investors in this period were **not compensated adequately for the level of volatility they endured**.

#### E. Beta Regression Lines (AAPL/MSFT vs S&P 500)



Using regression analysis:

- **Beta (Slope)** = How much the stock moves when the market moves.

- **R<sup>2</sup> (Coefficient of Determination)** = How well the market explains the stock's movement.

This regression chart analyzes the daily return relationship between Apple Inc. (AAPL) and the S&P 500 index (^GSPC), providing valuable insights into AAPL's market sensitivity.

- **Regression Equation & Interpretation**

The trendline equation in the chart is:  **$y = 0.9324x - 0.0004$**

This implies that for every 1% daily return movement in the S&P 500, Apple's stock tends to move approximately 0.93% in the same direction.

- **Beta Value (0.9324):**

This value represents Apple's beta coefficient, a measure of its systematic risk or volatility about the market. A beta of 0.93 suggests that **AAPL is slightly less volatile than the market.**

If the market increases by 1%, AAPL typically increases by 0.93%.

If the market falls by 1%, AAPL tends to decrease by about 0.93%.

- **Intercept (-0.0004):**

This is close to zero, indicating that **there's minimal unexplained bias** or average deviation when the market return is zero.

- **R<sup>2</sup> Value (0.9361)**

The coefficient of determination,  $R^2 = 0.9361$ , signifies that approximately 93.61% of AAPL's daily return variability can be explained by the S&P 500's daily movements. This reflects a very strong correlation between AAPL and the market index, showcasing that AAPL's stock performance is heavily driven by overall market trends.

The scatter plot displays a tight cluster of data points around the regression line, visually confirming a strong linear relationship.

The absence of extreme outliers suggests stable return behavior without significant anomalies during the observed period.

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#### F. Correlation Table

|       | AAPL     | MSFT     | ^GSPC    |
|-------|----------|----------|----------|
| AAPL  | 1        | 0.94678  | 0.967514 |
| MSFT  | 0.94678  | 1        | 0.970638 |
| ^GSPC | 0.967514 | 0.970638 | 1        |

- The correlation matrix reveals a **strong positive relationship** between the daily returns of AAPL, MSFT, and the S&P 500 index (^GSPC). Specifically, AAPL and MSFT exhibit a **correlation of 0.9468**, suggesting that the two stocks tend to move in the same direction with high consistency. Both AAPL and MSFT also show even **stronger correlations** with the S&P 500—0.9675 and 0.9706, respectively, indicating that their **price movements are highly aligned with overall market trends**.
- This suggests that these tech giants are not only closely linked to each other but are also **major contributors to broader market performance**. Such high correlations imply that diversification benefits across these assets may be limited in a portfolio context, as they **tend to respond similarly to market-wide factors**.

## 8. Summary Table (Big Six Style)

| Annual Return, Volatility, Beta and R <sup>2</sup> (Big Six Style) |                   |                       |      |                |
|--------------------------------------------------------------------|-------------------|-----------------------|------|----------------|
|                                                                    | AAPL              |                       | MSFT |                |
|                                                                    | Annual Return (%) | Annual Volatility (%) | Beta | R <sup>2</sup> |
|                                                                    | 21.59             | 23.62                 | 1.08 | 1.08           |
| MSFT                                                               | 21.65             | 23.17                 | 0.43 | 0.44           |
|                                                                    | 13.63             | 14.24                 |      |                |
| S&P500                                                             |                   |                       |      |                |

- **Performance:** Both **AAPL** and **MSFT** significantly outperformed the S&P 500 in terms of annual return, demonstrating strong investor confidence and growth potential.
- **Risk (Volatility):** Higher than the market average (S&P 500 at 14.24%) — reflects **more aggressive price swings**, typical for growth and tech stocks.
- **Beta (1.08):** Indicates that both stocks are **slightly more sensitive** to market movements. When the S&P 500 moves by 1%, these stocks tend to move ~1.08%.
- **R<sup>2</sup> Values:** Around 0.43–0.44, meaning **only 43–44% of the stock's movement** is explained by the market, indicating **strong company-specific factors** driving performance.

## Business Implications

Despite early strong performance across AAPL, MSFT, and the S&P 500, Q1 2025 signals a possible market correction or macroeconomic shock. MSFT shows stronger resilience compared to AAPL, maintaining higher adjusted closes throughout. A business investor should be cautious of overall market signals while noting MSFT's relative strength for future positions.

## 9. Executive Summary

This report presents a comprehensive investment analysis of Apple Inc. (AAPL) and Microsoft Corporation (MSFT), benchmarking their performance against the S&P 500 index. Using

historical stock data, we evaluated key financial indicators such as daily and annual returns, volatility, beta, and  $R^2$  to understand the risk-return profile of each asset.

The analysis reveals that both AAPL and MSFT consistently outperform the S&P 500 in terms of average returns, but with varying levels of volatility and market sensitivity. While AAPL exhibits higher volatility and stronger price swings, MSFT demonstrates more stability and a balanced beta, indicating lower risk in relation to market fluctuations.

Visualizations generated using Excel provide a clear understanding of trends and comparisons, while Python scripts were utilized for deeper statistical analysis and validation. The combination of tools ensures both accuracy and clarity in investment insights.

This report is designed to assist stakeholders and potential investors in making informed decisions regarding portfolio diversification, asset selection, and risk management.

## 10. Conclusion

This investment analysis evaluates the performance of **Apple Inc. (AAPL)** and **Microsoft Corp. (MSFT)** against the **S&P 500 index** across a multi-year period, using key financial metrics and statistical models.

Both AAPL and MSFT have demonstrated **steady financial growth**, with **revenue and net profit increasing over time**, especially in 2022, which marked their **highest profit margins**. These improvements reflect **strong operational efficiency and cost management**.

Our analysis indicates that while **both stocks exhibited higher volatility than the S&P 500**, they also delivered **higher annual returns**, pointing to **greater potential for long-term growth**. However, during the selected analysis timeframe, **negative short-term returns** were observed, highlighting **market cycles and economic uncertainty**.

- The **beta coefficients (~1.08)** show both stocks are moderately sensitive to market fluctuations, while the  **$R^2$  values (~0.43–0.44)** confirm that a large portion of price movements are driven by **company-specific factors**.

- The **correlation matrix and heatmap** revealed strong relationships between the two stocks and the broader index, while **volume outlier analysis** showed spikes in trading activity around **earnings, macroeconomic events, and triple witching days**.
- Visualization tools such as **trend lines, bar plots, beta regression, rolling volatility, histograms, and scatter plots** provided additional clarity on each stock's **risk-return profile** and market behavior.

In summary, AAPL and MSFT show **strong fundamentals** and growth potential, yet require **risk-aware investment strategies** due to their higher volatility and market sensitivity.

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## 11. Recommendations

Based on the insights gathered, the following strategic actions are recommended:

### a. Diversify for Risk Mitigation

- Due to their **higher volatility**, balance AAPL and MSFT with low-beta assets or index funds like the **S&P 500** to maintain a stable portfolio.

### b. Monitor Key Events

- Pay attention to **quarterly earnings, product launches, and macroeconomic news**, as these frequently trigger high volume and volatility days.

### c. Plan for Long-Term Gains

- Despite short-term dips, both stocks have **strong fundamentals** and remain good candidates for **long-term growth**. Use **dollar-cost averaging** for steady entry points.

### d. Manage Risk Proactively

- Use **stop-loss orders** or hedging strategies to protect against market swings.

- Reassess **rolling volatility** and **beta trends** quarterly to stay aligned with portfolio goals.

#### e. Stay Informed

- Since company-specific news plays a big role, actively monitor **innovation announcements**, **leadership updates**, and **competitive positioning**.
-